



Agricultural Productivity Analysis Dashboard

Overview of rainfall, crop yields (Rice, Wheat, Kharif), regional comparisons, temporal trends, and actionable insights from a 52-year dataset covering 18 states.

Key Metrics Snapshot

Total Rainfall

877.68K mm

Average Rainfall

937.69 mm

Avg Rice Yield

1.61K kg/ha (max 4.66K)

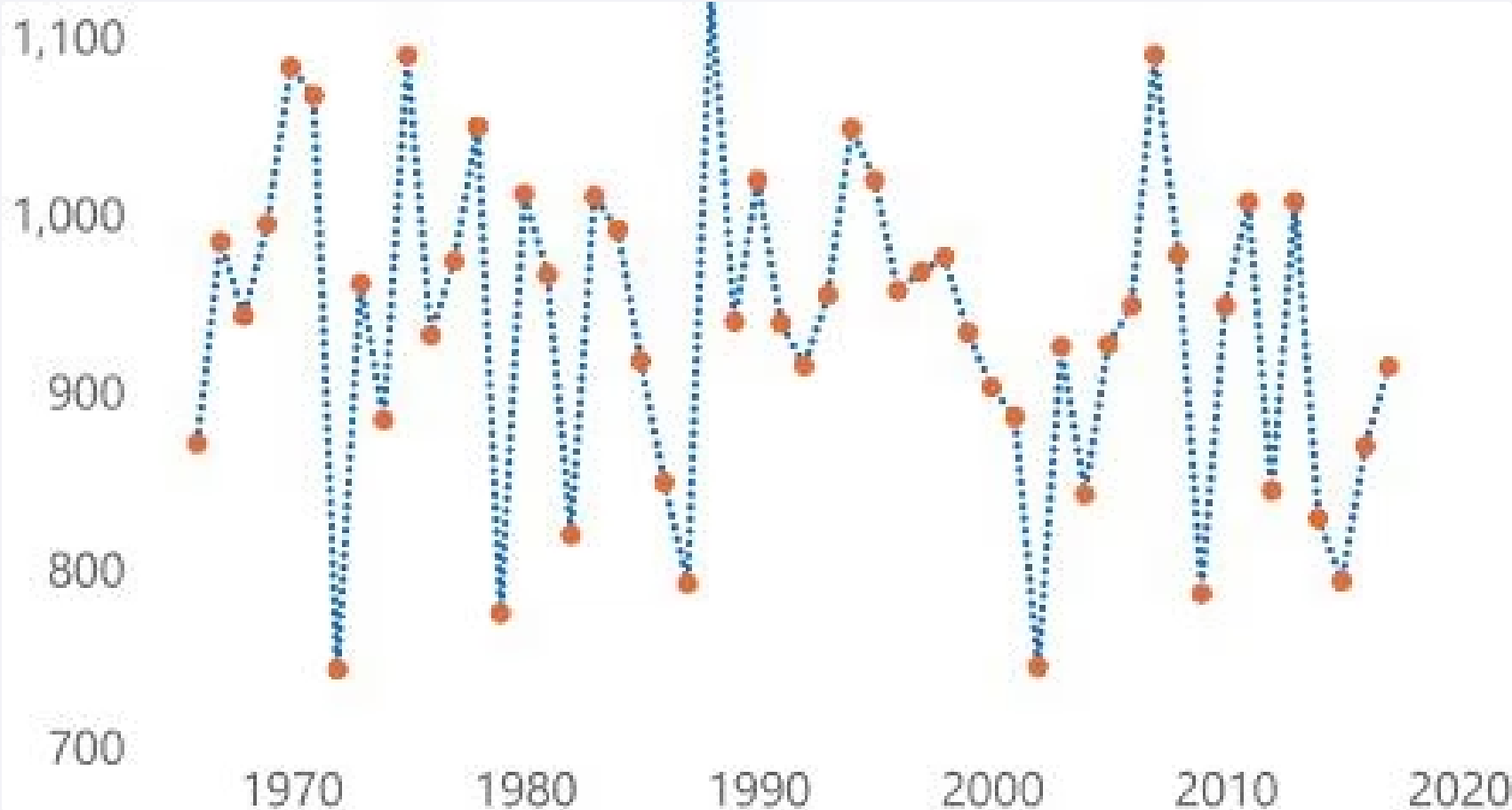
Avg Wheat Yield

1.45K kg/ha (max 5.09K)

Avg Kharif Yield

441.96 kg/ha (max 2.64K)

Rainfall Trend (1970–2020)



Average annual rainfall shows strong inter-annual variability with notable peaks (circa 1975, 1985, 1995) and troughs (circa 1978, 1982, 2002), indicating an inconsistent monsoon pattern that affects productivity.

Total Rainfall Trend Over Time by State (1966–2017)

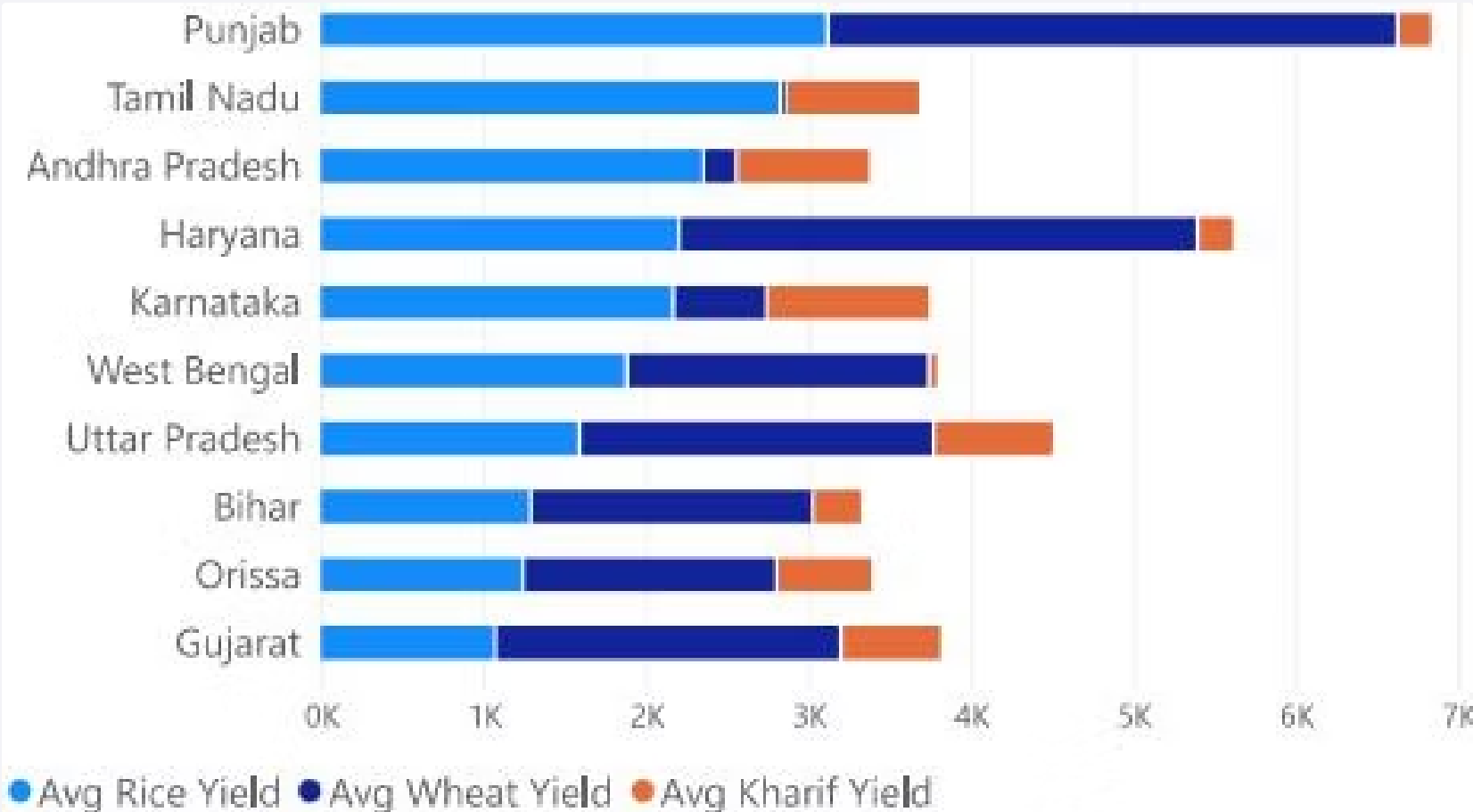
Total Rainfall by Year and State Name

Legend:

- Andhra Pradesh
- Assam
- Bihar
- Gujarat
- Haryana
- Himachal Pradesh
- Jharkhand
- Karnataka
- Kerala
- Madhya Pradesh
- Maharashtra
- Orissa
- Punjab
- Rajasthan
- Tamil Nadu
- Uttar Pradesh
- Uttarakhand
- West Bengal

- Regions matter: policy and irrigation must be region-specific.

Top States by Average Crop Yield



Punjab

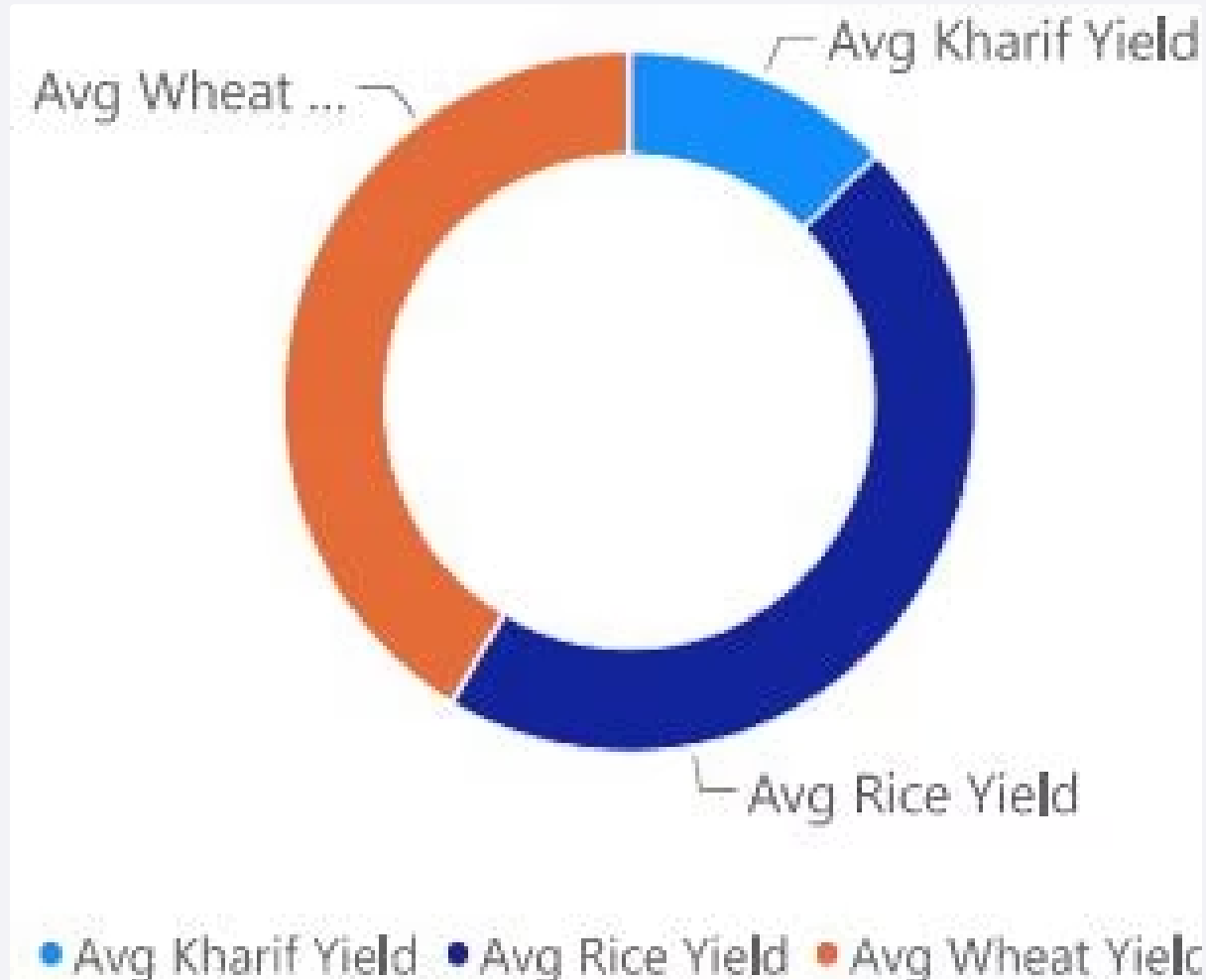
Highest yields (Rice ~3.2K, Wheat ~6.5K).
Strong performance despite moderate rainfall
—suggests robust irrigation & practices.



Tamil Nadu & Andhra Pradesh

High rice yields (3.0K; 2.5K). Regional
strengths differ by crop mix and water
regimes.

Crop Yield Composition

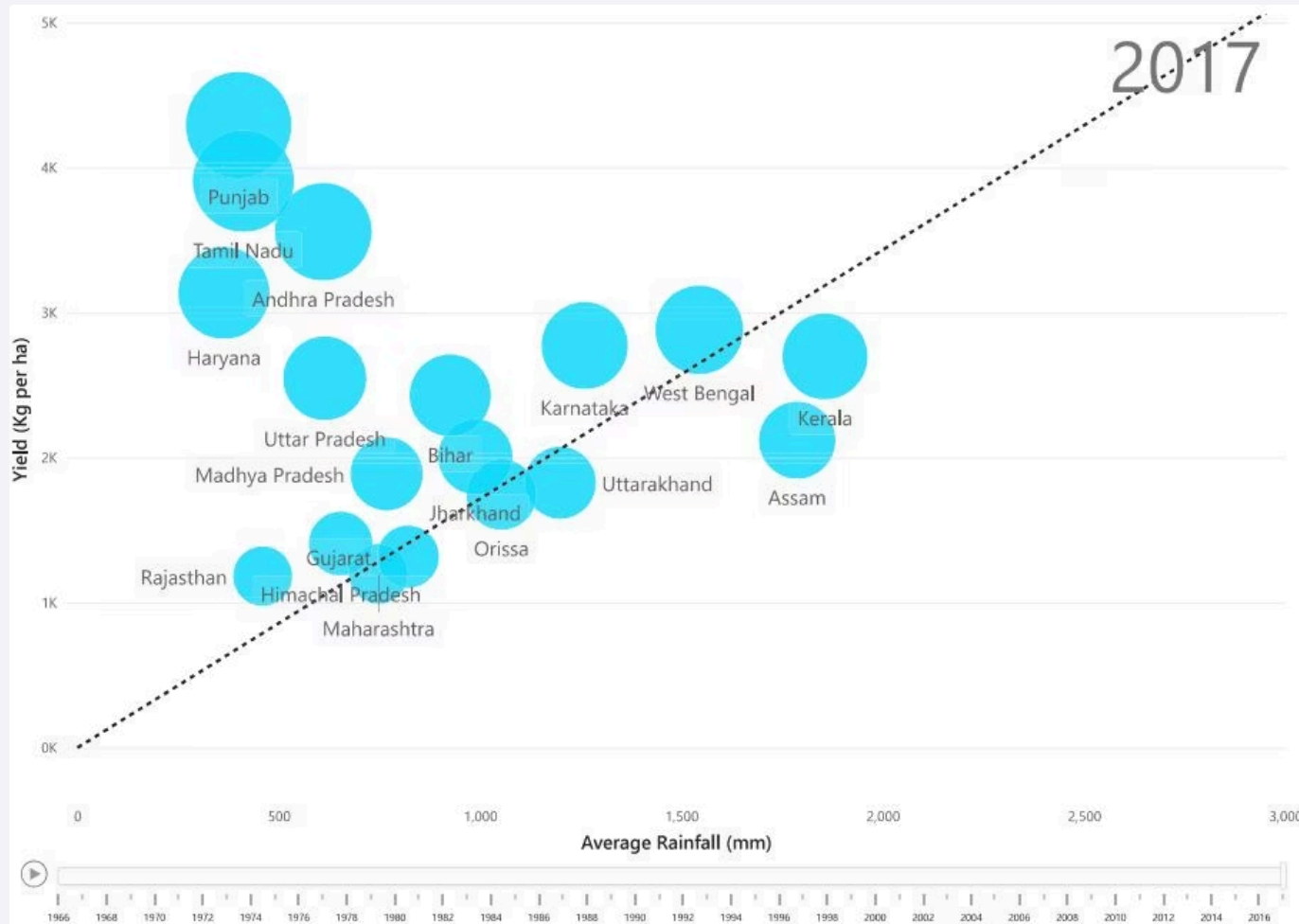


Distribution of average yields (approx.):

- Avg Rice Yield — 1.61K (~45%)
- Avg Wheat Yield — 1.45K (~5%)
- Avg Kharif Yield — 441.96 (~50%)

Rice and wheat dominate total yield; Kharif contributes a distinct but smaller share depending on region.

Rainfall vs Rice Yield (State-level)



Rice shows a moderate positive correlation with rainfall: higher yields generally occur where rainfall is adequate but not exclusively—Punjab achieves top rice yields with moderate rainfall, indicating critical role of irrigation.

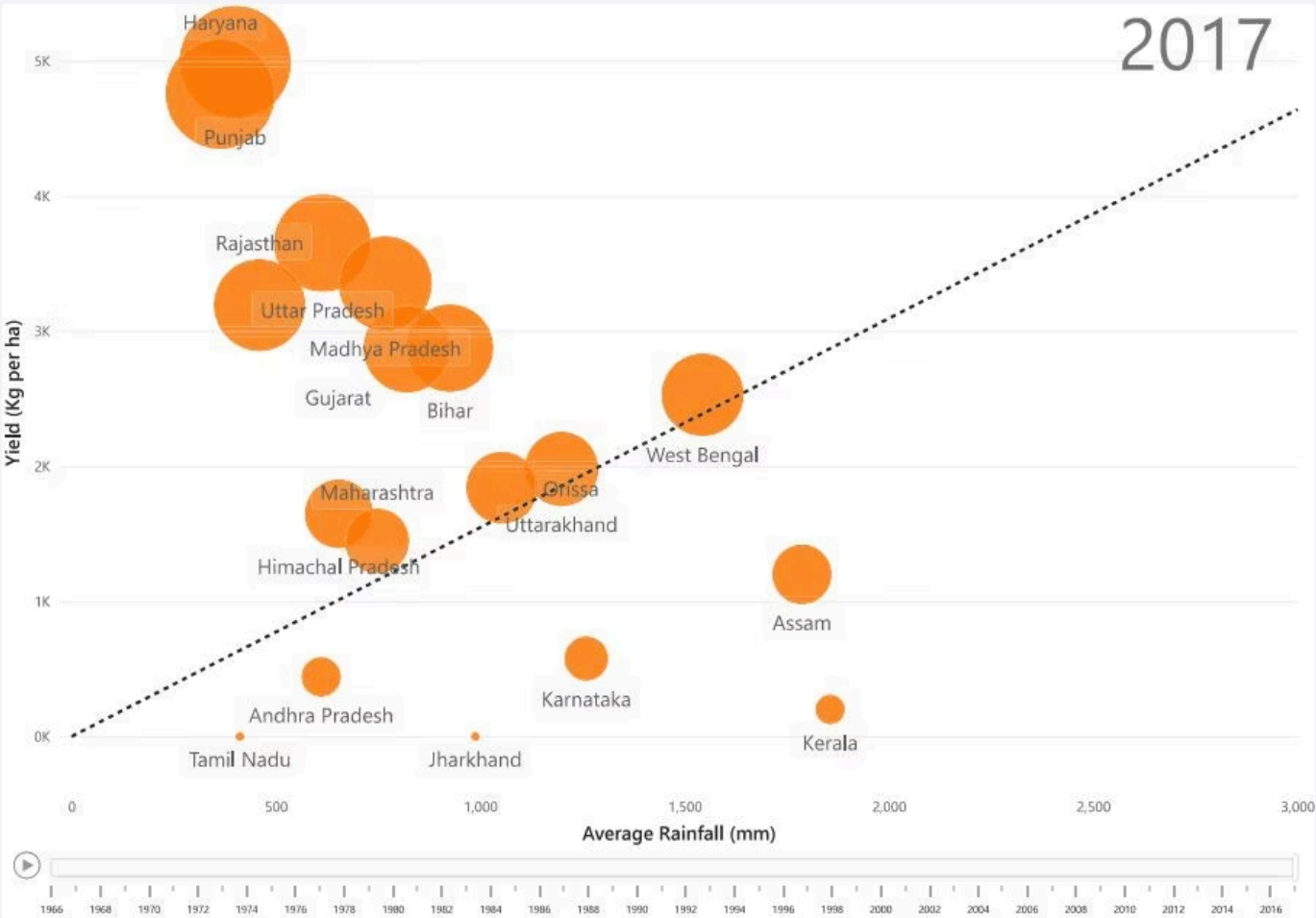
Notable Points

Highest rainfall: Kerala, Assam. Highest rice yield: Punjab (~4.66K).

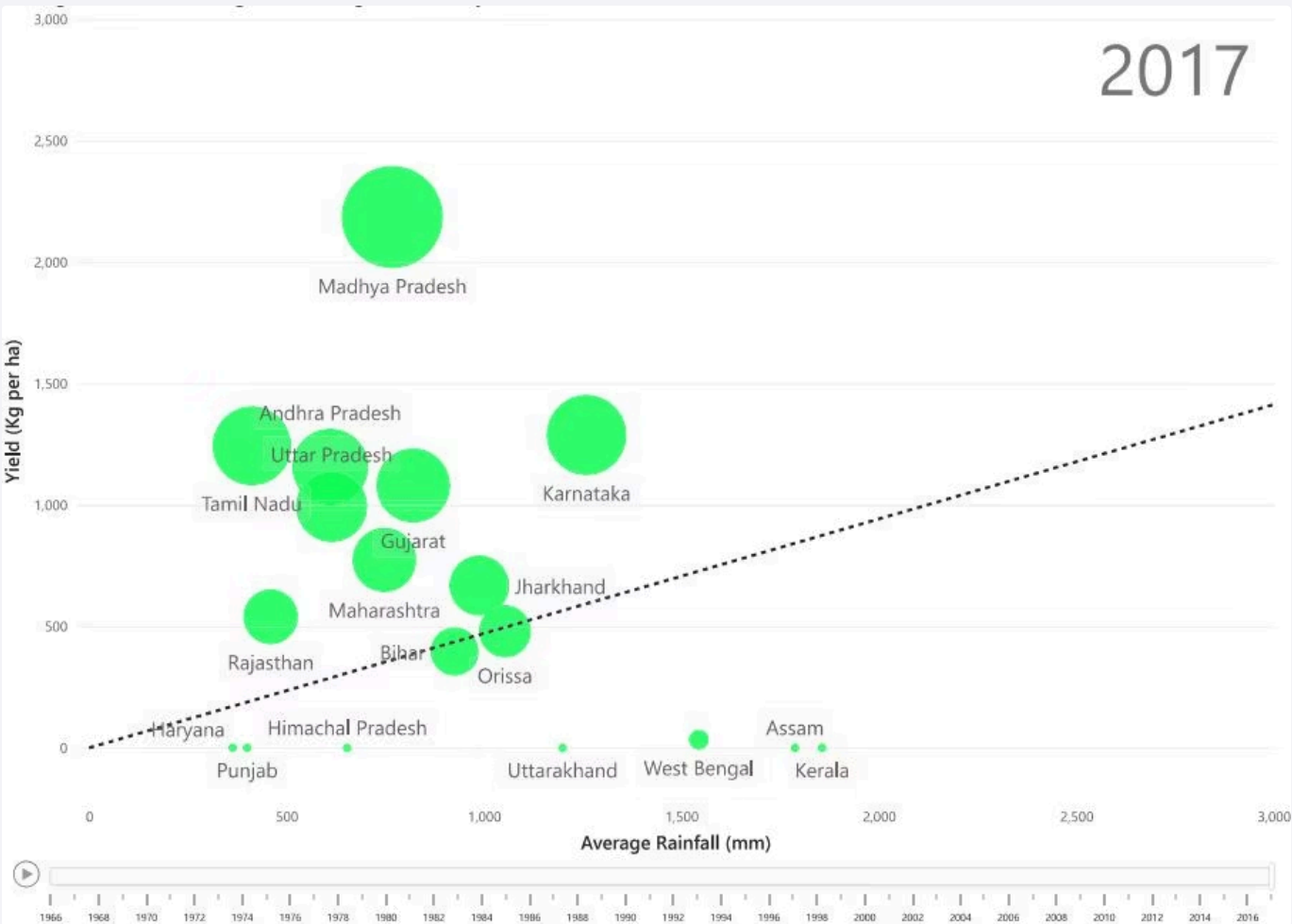
Implication

Irrigation and practices can offset lower rainfall.

Rainfall vs Wheat & Kharif Yields

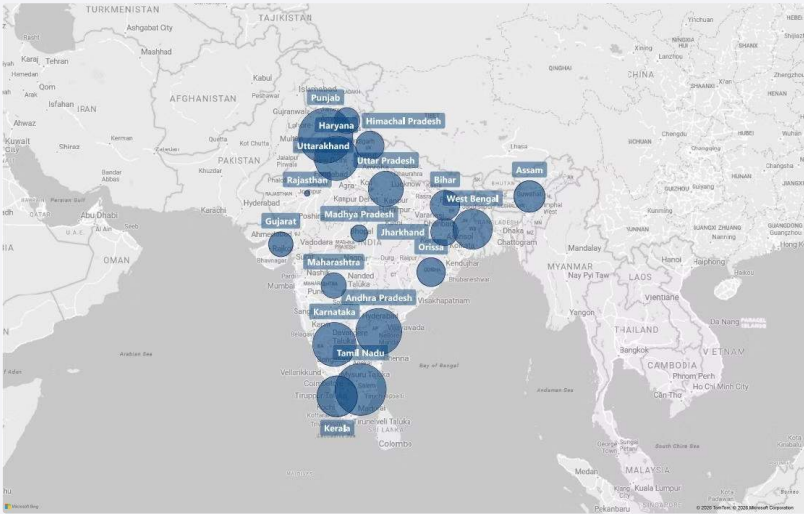


Wheat: weaker dependence on rainfall; top yields in drier, well-irrigated zones (Punjab, Haryana). Kharif: strong dependence on monsoon—regions with stable monsoon see better Kharif returns.



- Wheat peaks in low-to-moderate rainfall zones (500–1,500 mm).
- Kharif performance closely tracks monsoon stability; Madhya Pradesh and parts of Karnataka lead Kharif yields.

Spatial Overview: Productivity Map & Drill-down



Map highlights state-level rainfall and yield. Drill-down treemap (example year 2007, low rainfall) shows selected yields: Punjab (Rice 3,908.31; Wheat 4,447.19) and other state contributions.

- Treemap reveals how specific years and rainfall categories concentrate production in a few high-performing states.

Insights & Recommendations

Rainfall variability

Monsoon is inconsistent across years and states; variability directly impacts Kharif and rice production.

Rice sensitivity

Rice yield is closely related to rainfall but can be boosted by irrigation—Punjab demonstrates this.

Wheat resilience

Wheat performs well in drier, irrigated zones; yields depend more on infrastructure and practices than on high rainfall.

Actionable recommendations

Invest in targeted irrigation, region-specific agronomy, and monsoon-adaptive cropping to stabilize yields and close regional gaps.



Next steps: prioritize irrigation in moderate-rainfall high-yield states, strengthen water storage in high-variance regions, and deploy tailored extension services to translate rainfall into reliable production.